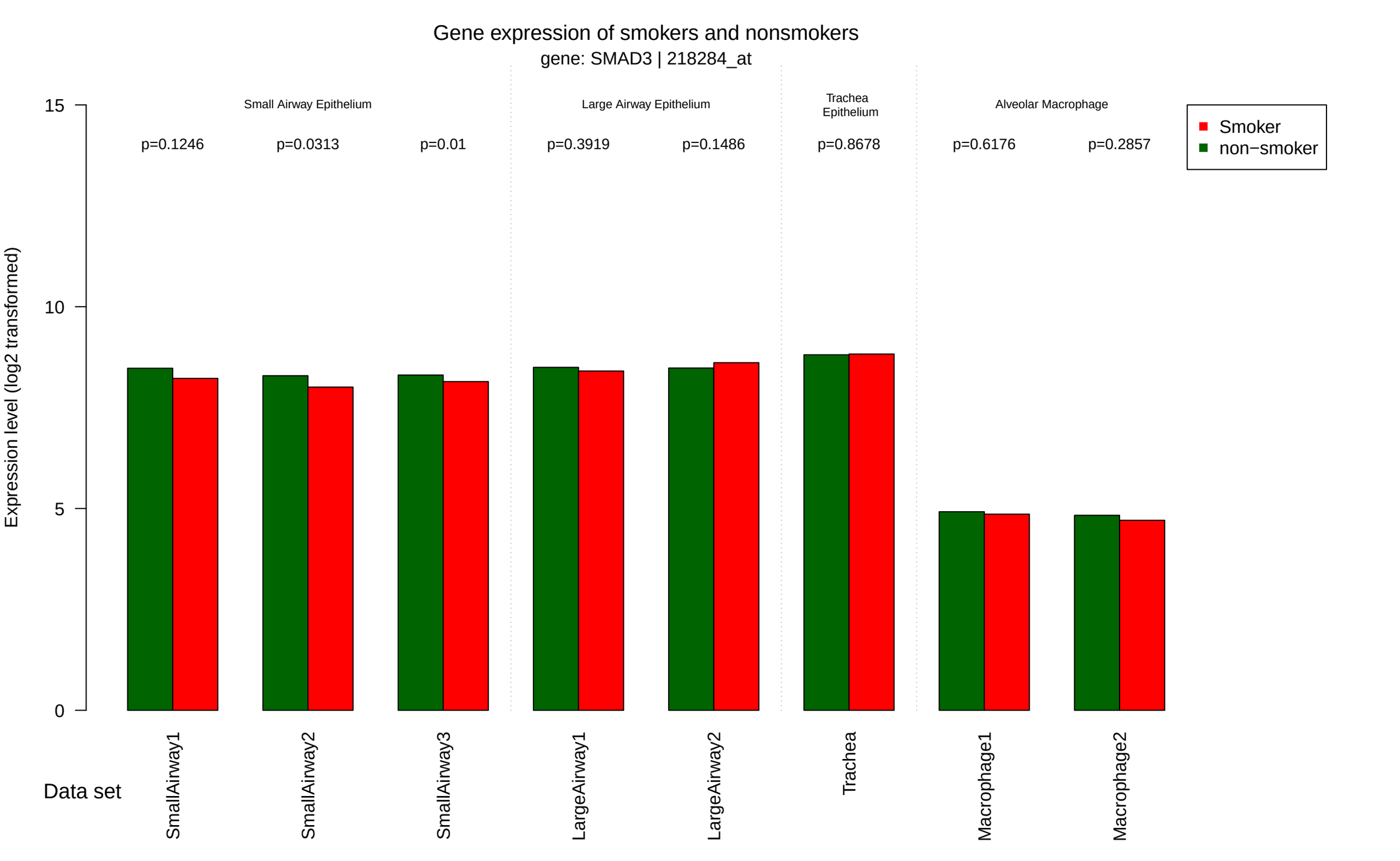
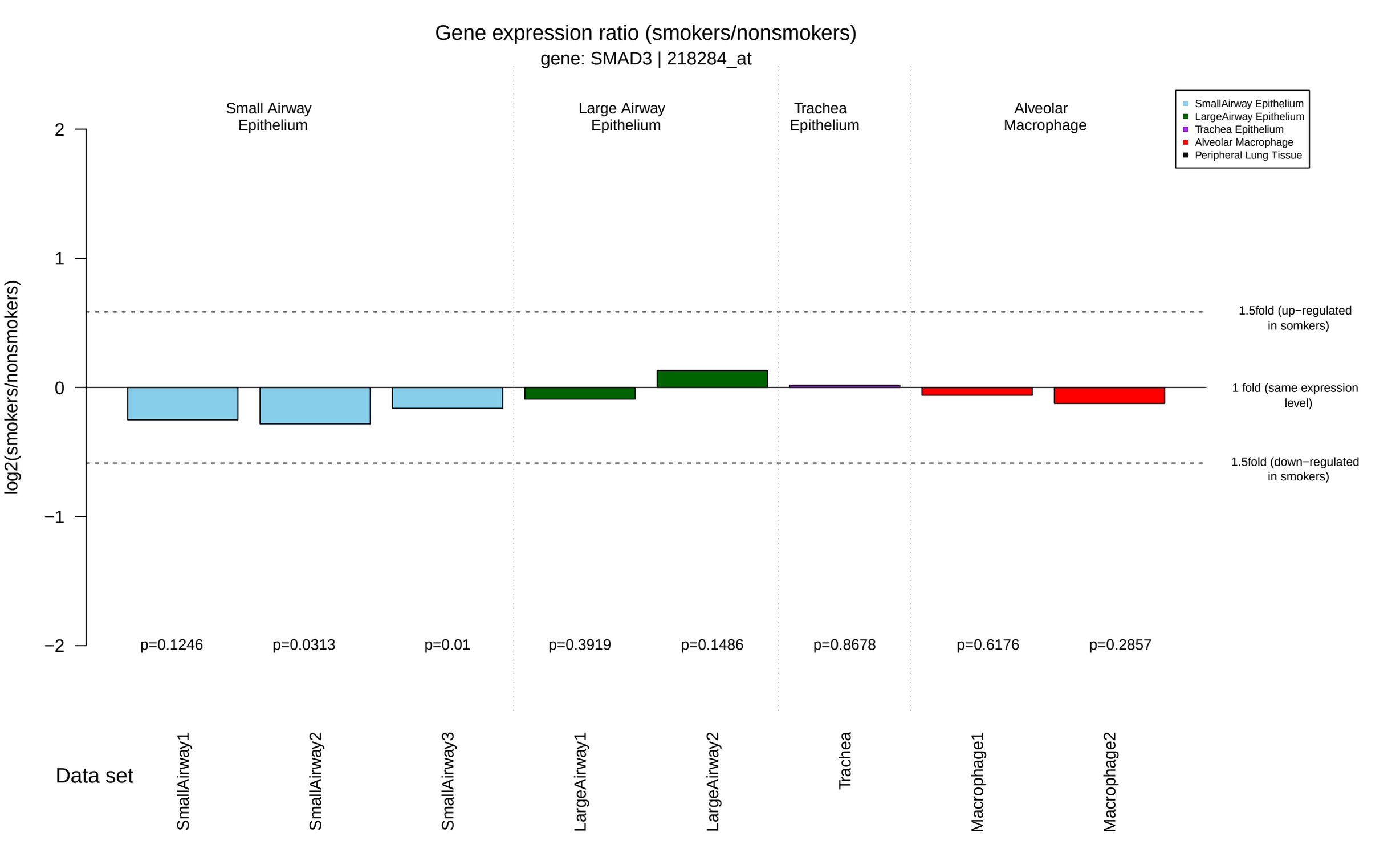
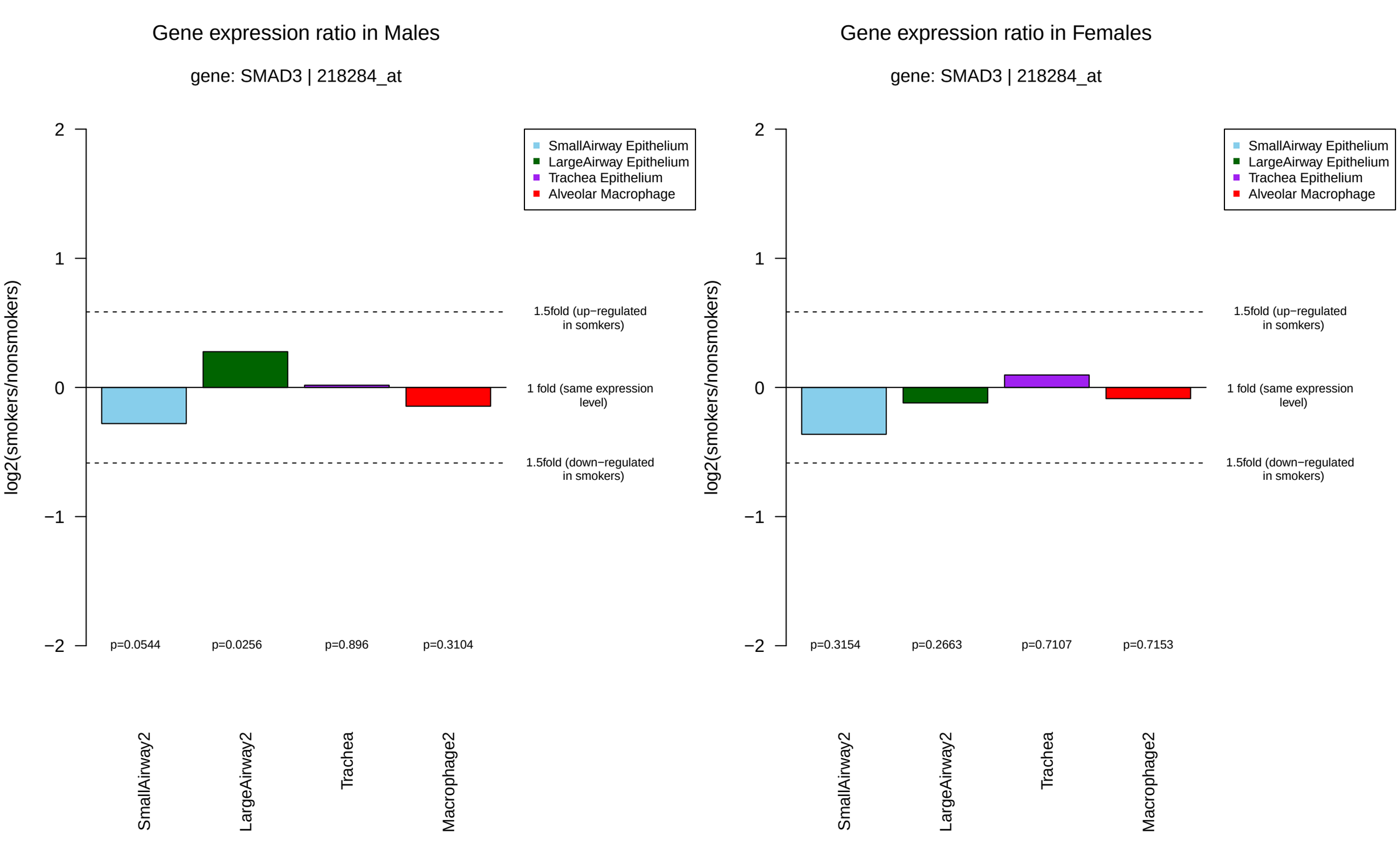
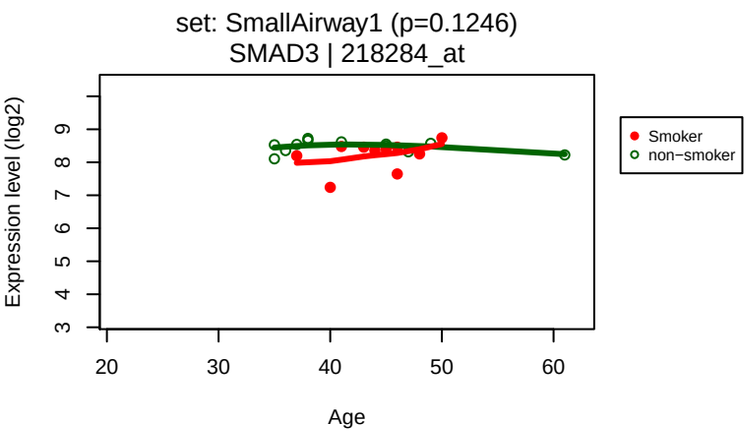
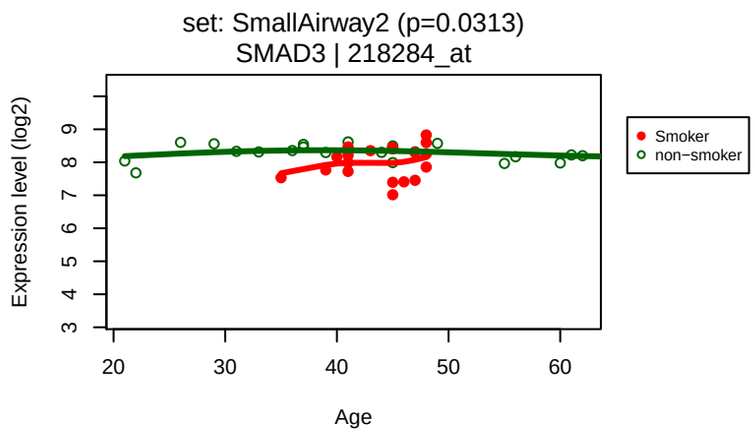
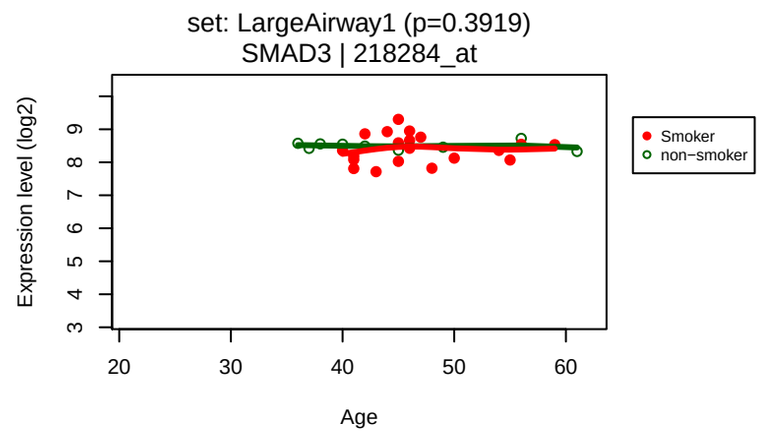

Gene expression of smokers and nonsmokers
gene: SMAD3 | 218284_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1246
p=0.0313
p=0.01
p=0.3919
p=0.1486
p=0.8678
p=0.6176
p=0.2857
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

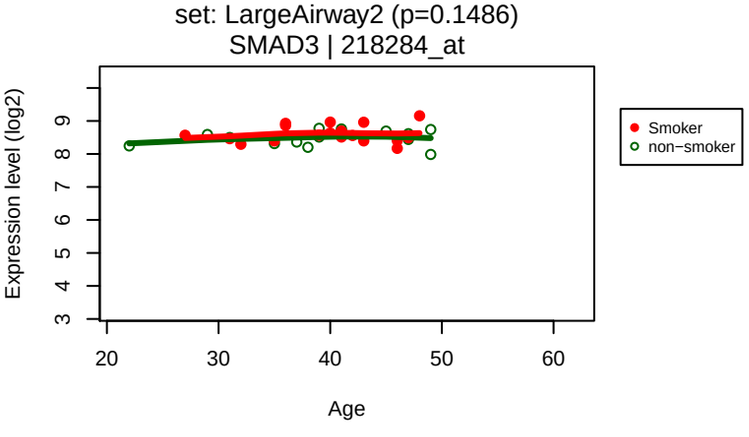

Gene expression ratio (smokers/nonsmokers)
gene: SMAD3 | 218284_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1246
p=0.0313
p=0.01
p=0.3919
p=0.1486
p=0.8678
p=0.6176
p=0.2857
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

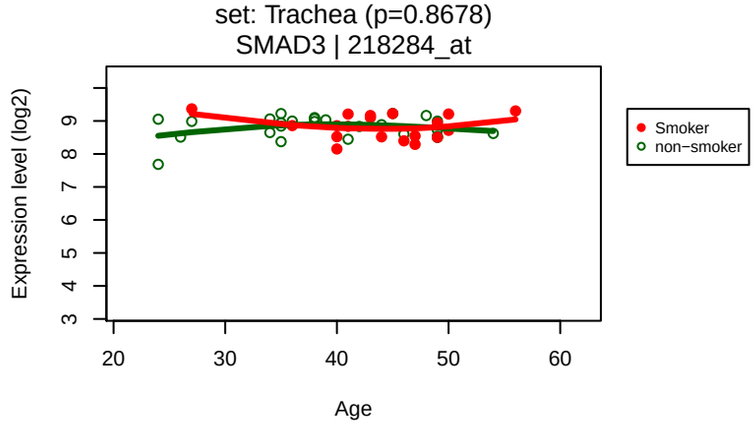

Gene expression ratio in Males
gene: SMAD3 | 218284_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0544
p=0.0256
p=0.896
p=0.3104
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD3 | 218284_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3154
p=0.2663
p=0.7107
p=0.7153
SmallAirway2
LargeAirway2
Trachea
Macrophage2

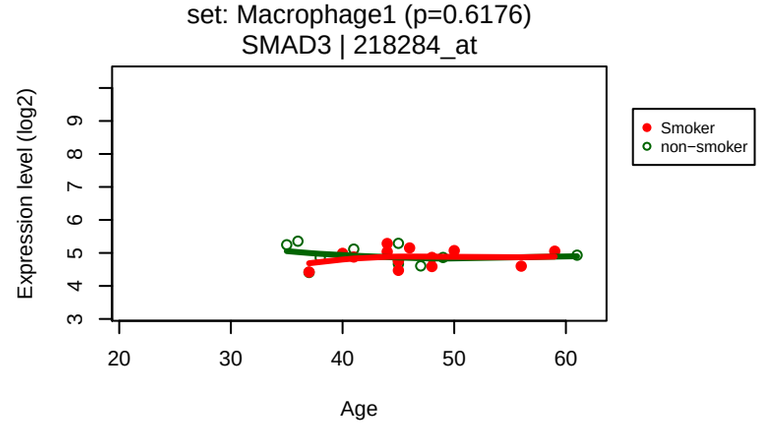

set: SmallAirway1 (p=0.1246)
SMAD3 | 218284_at
Expression level (log2)
Age
Smoker
non−smoker

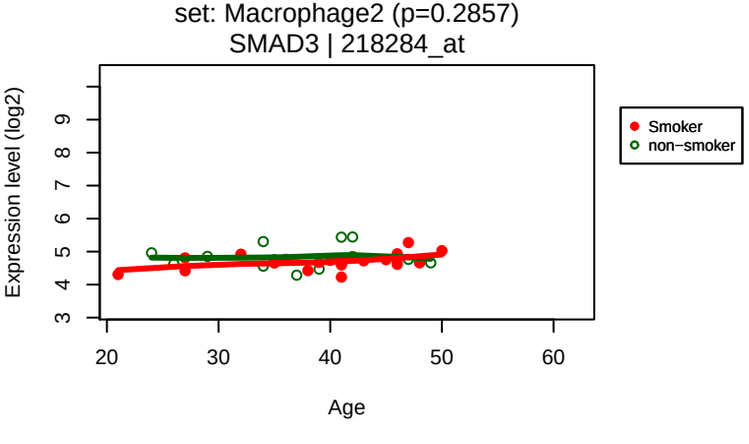

set: SmallAirway2 (p=0.0313)
SMAD3 | 218284_at
Expression level (log2)
Age
Smoker
non−smoker

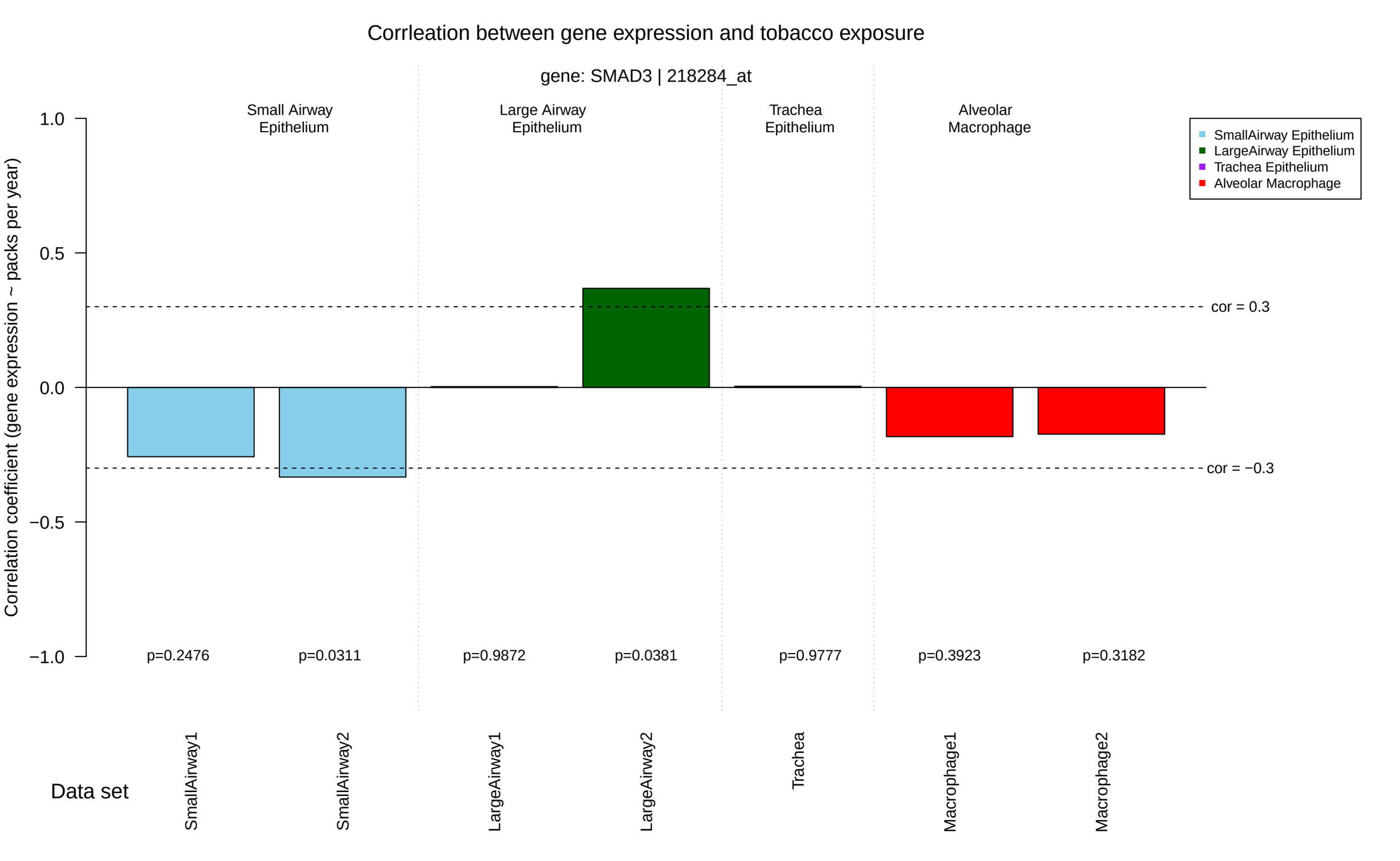

set: LargeAirway1 (p=0.3919)
SMAD3 | 218284_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1486)
SMAD3 | 218284_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8678)
SMAD3 | 218284_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6176)
SMAD3 | 218284_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2857)
SMAD3 | 218284_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD3 | 218284_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2476
p=0.0311
p=0.9872
p=0.0381
p=0.9777
p=0.3923
p=0.3182
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2